CrewAI RevenueOps Intelligence

An evidence-led financial operations workspace where deterministic software protects money, specialist agents investigate exceptions, and people retain approval authority.

Full-stack engineering | Multi-agent orchestration | Financial controls

EXECUTIVE SUMMARY

I engineered a connected accounts-receivable and payment-reconciliation system with a React operational console, authenticated FastAPI services, PostgreSQL persistence, and a CrewAI investigation layer. The design deliberately separates financial truth from model interpretation: exact values, reconciliation proposals, locking and ledger changes are deterministic. Agents can inspect scoped evidence and propose explanations; only an authorised reviewer can approve an allocation.

What the application supports

01 **Ingest & reconcile**

CSV preview, row validation, source provenance, duplicate protection, multi-currency precision and conservative reconciliation proposals.

02 **Investigate exceptions**

Four specialist CrewAI roles, typed Flows, scoped read-only tools, structured evidence citations and saved investigation outcomes.

03 **Review & account**

Role-controlled decisions, immutable allocations, dispute handling, auditable changes and draft collections without external sending.

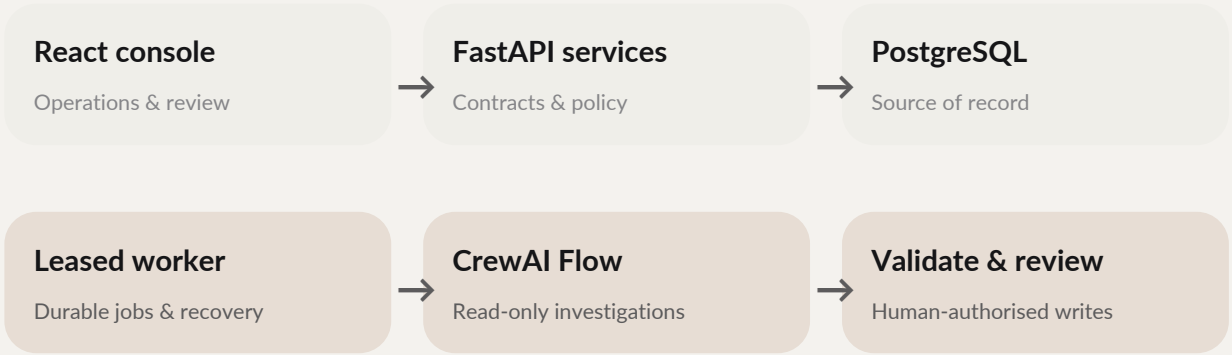
04 **Operate & recover**

Leased PostgreSQL workers, checkpoints, fenced retries, correlation records, evaluations and independently gated release tooling.

Separate the decisions from the authority.

The architecture treats language-model output as advisory, persisted evidence as inspectable, and financial mutations as strictly controlled application operations.

RESPONSIBILITY MAP



The worker checkpoints results in PostgreSQL. Model runs do not hold an open database transaction; stale worker leases are fenced. The agent has no posting, messaging, SQL, shell or arbitrary network tool.

Four engineering decisions

Exact amounts stay exact

Decimal accounting; supported currency precision; derived balances from immutable allocations; no FX or mutable balance truth.

Every allocation is reviewed

Fresh organisation-scoped validation under locks, reviewer permissions and atomic audit/idempotency records.

Agents read, never post

Conditional specialist crews use one scoped evidence tool; case identity, citations and outputs face guardrails.

Failure is part of state

PostgreSQL leases, checkpoints, bounded run times, explicit errors and resumable completed inference.

Measured, not assumed.

The engineering is verifiable, and the model evaluation is intentionally reported alongside its limitations.

100
Backend & release tests
PostgreSQL and release suite

5
Browser workflows
Hosted end-to-end checks

13/13
Financial regression cases
Small deterministic labelled set

5/5
Hosted CI jobs
Latest documented interface stage

MODEL EVALUATION

Frozen ambiguous-payment cases, local llama3.2:3b. Exact decision accuracy - not financial posting.

Fixed cases		Sealed holdouts	
CrewAI	0 / 4	CrewAI	1 / 7
Baseline	1 / 4	Baseline	4 / 7

Across the eleven evaluated cases, nine failed typed output. All 24 read-tool calls succeeded, but that did not improve exact investigation outcomes. No allocations were posted by a model.

Release and operational boundary

Verified

Local application workflows; PostgreSQL integration and concurrency checks; frontend unit tests; five hosted browser workflows; five hosted CI jobs and container builds.

Prepared, not deployed

Protected staging promotion, release manifests and rollback checks exist as reviewed automation. No staging activation, registry publishing or production rollout occurred.

Scope: Synthetic business records; no live banking, payment execution, external email, FX or SSO. Four reviewed transitive Chroma server advisories remained at the source audit; the relevant server functionality was not enabled.

SOURCE REPOSITORY

https://github.com/keenosmith-del/crewai-revenueops

Source revision f7270742ec3c / 26 Oct 2026